

35. Build a vector database using FAISS or ChromaDB and perform similarity-based document retrieval.

Code:

```
from sentence_transformers import SentenceTransformer
import faiss
import numpy as np

# Load embedding model
model = SentenceTransformer("all-MiniLM-L6-v2")

# Documents
documents = [
    "Python is a popular programming language.",
    "Machine learning allows computers to learn from data.",
    "Artificial intelligence is used to build smart systems.",
    "Natural language processing deals with human language.",
    "Python is widely used for data science.",
    "Deep learning uses neural networks to learn complex patterns.",
    "Computer networks allow computers to communicate with each other."
]

# Generate document embeddings
embeddings = model.encode(documents)

# Convert embeddings to NumPy array
embeddings = np.array(embeddings).astype("float32")
```

```
# Get embedding dimension
dimension = embeddings.shape[1]

# Create FAISS index
index = faiss.IndexFlatL2(dimension)

# Add document embeddings to FAISS
index.add(embeddings)

print("Number of documents stored in FAISS:", index.ntotal)

# Get query from user
query = input("\nEnter your search query: ")

# Generate query embedding
query_embedding = model.encode([query])
query_embedding = np.array(query_embedding).astype("float32")

# Number of results to retrieve
k = 3

# Search FAISS database
distances, indices = index.search(query_embedding, k)

# Display results
print("\nTOP", k, "SIMILAR DOCUMENTS")
print("-----")

for i in range(k):
```

```
document_index = indices[0][i]
```

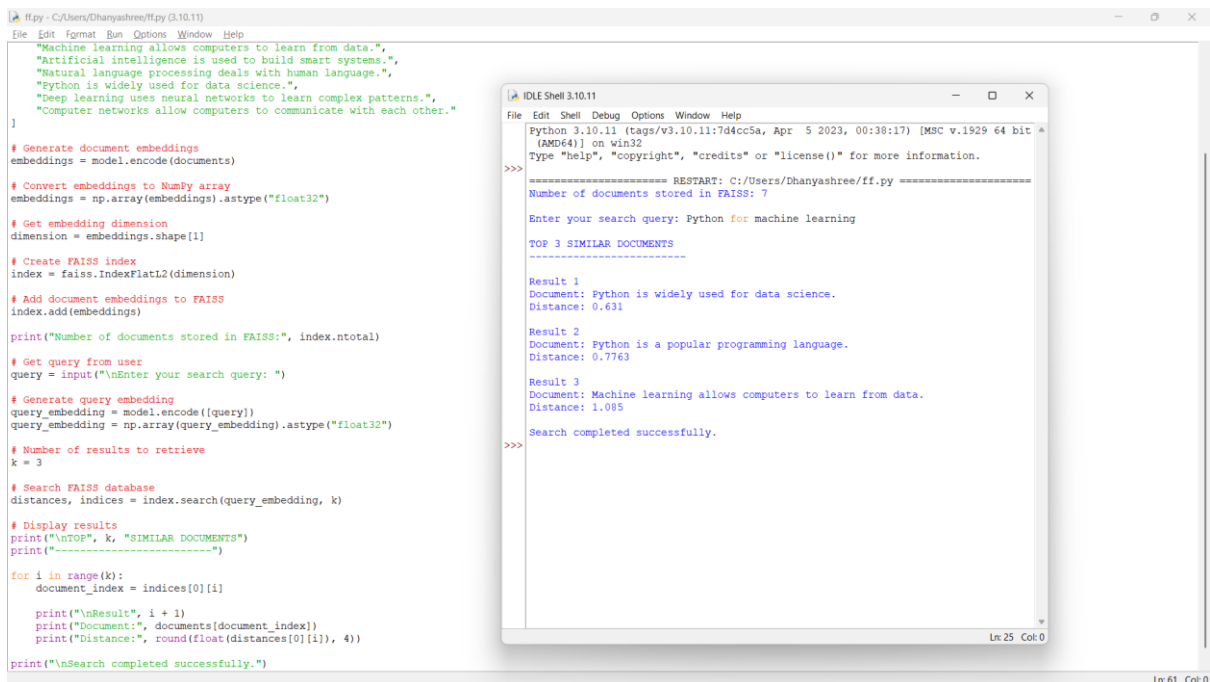
```
print("\nResult", i + 1)
```

```
print("Document:", documents[document_index])
```

```
print("Distance:", round(float(distances[0][i]), 4))
```

```
print("\nSearch completed successfully.")
```

output:



The screenshot shows a Python script in the IDLE editor and its execution output in the IDLE Shell. The script uses FAISS for document similarity search. It defines a list of documents, generates embeddings, creates a FAISS index, and searches for documents similar to a query. The output shows the top 3 similar documents and their distances.

```
ffpy - C:/Users/Dhanyashree/ffpy (3.10.11)
File Edit Format Run Options Window Help
"Machine learning allows computers to learn from data.",
"Artificial intelligence is used to build smart systems.",
"Natural language processing deals with human language.",
"Python is widely used for data science.",
"Deep learning uses neural networks to learn complex patterns.",
"Computer networks allow computers to communicate with each other."
}

# Generate document embeddings
embeddings = model.encode(documents)

# Convert embeddings to NumPy array
embeddings = np.array(embeddings).astype("float32")

# Get embedding dimension
dimension = embeddings.shape[1]

# Create FAISS index
index = faiss.IndexFlatL2(dimension)

# Add document embeddings to FAISS
index.add(embeddings)

print("Number of documents stored in FAISS:", index.ntotal)

# Get query from user
query = input("\nEnter your search query: ")

# Generate query embedding
query_embedding = model.encode([query])
query_embedding = np.array(query_embedding).astype("float32")

# Number of results to retrieve
k = 3

# Search FAISS database
distances, indices = index.search(query_embedding, k)

# Display results
print("\nTOP", k, "SIMILAR DOCUMENTS")
print("-----")

for i in range(k):
    document_index = indices[0][i]
    print("\nResult", i + 1)
    print("Document:", documents[document_index])
    print("Distance:", round(float(distances[0][i]), 4))

print("\nSearch completed successfully.")

IDLE Shell 3.10.11
File Edit Shell Debug Options Window Help
Python 3.10.11 (tags/v3.10.11:7d4cc5a, Apr 5 2023, 00:38:17) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/Dhanyashree/ffpy =====
Number of documents stored in FAISS: 7

Enter your search query: Python for machine learning

TOP 3 SIMILAR DOCUMENTS
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Result 1
Document: Python is widely used for data science.
Distance: 0.631

Result 2
Document: Python is a popular programming language.
Distance: 0.7763

Result 3
Document: Machine learning allows computers to learn from data.
Distance: 1.085

Search completed successfully.
>>>
```